

HDOC 15

Name: Keshav Mittal SAP ID: 590032934

Batch: 16

1. Algorithm

1. Start
2. Declare variables for choice, num, temp, rem, all_even, and all_prime.
3. Display the menu options to the user.
4. Read the user's choice.
5. If choice is 1 or 2, prompt the user to input a positive integer num. Validate that $\text{num} > 0$.
6. Use a switch statement based on the value of choice.
7. Case 1 (Check if all digits are even):
 - Set all_even = 1 and temp = num.
 - Loop while (temp > 0):
 - Calculate $\text{rem} = \text{temp} \% 10$.
 - If $\text{rem} \% 2 \neq 0$, set all_even = 0 and break the loop.
 - Update $\text{temp} = \text{temp} / 10$.
 - If all_even == 1, display "All digits are even". Else, display "All digits not even".
 - Break.
8. Case 2 (Check if all digits are prime):
 - Set all_prime = 1 and temp = num.
 - Loop while (temp > 0):
 - Calculate $\text{rem} = \text{temp} \% 10$.

- If rem is not 2, 3, 5, or 7, set all_prime = 0 and break the loop.
- Update temp = temp / 10.
- If all_prime == 1, display "All digits are prime". Else, display "All digits are not prime".
- Break.

9. Default Case: Display an "Invalid choice" error message.

10. Stop

2. Test Case Table

TC ID	Scenario Description	Inputs (choice, num)	Expected Output
TC-01	Valid inputs (All even digits)	Choice: 1 Num: 24680	Output: All digits are even
TC-02	Valid inputs (Mixed odd/even)	Choice: 1 Num: 12345	Output: All digits not even
TC-03	Valid inputs (All prime digits)	Choice: 2 Num: 2357	Output: All digits are prime
TC-04	Valid inputs (Mixed prime/composite)	Choice: 2 Num: 2379	Output: All digits are not prime
TC-05	Invalid negative number input	Choice: 1 Num: -200	Error: Please enter a strictly positive integer.
TC-06	Invalid menu choice	Choice: 4	Invalid choice. Please choose 1 or 2.

3. C Program Source Code

```
#include <stdio.h>

int main() {
    int choice;
    long long num, temp;
    int rem;
    int all_even, all_prime;

    // Display Menu
```

```

printf("--- Digit Operations Menu (HDOC 15) ---\n"); printf("1. Check whether
all digits are even\n"); printf("2. Check whether all digits are prime\n");
printf("Enter your choice (1 or 2): ");
scanf("%d", &choice);

// Validate Choice and Take Number Input if (choice == 1 ||
choice == 2) {
    printf("Enter a positive integer: "); scanf("%lld", &num);

    if (num <= 0) {
        printf("Error: Please enter a strictly positive integer.\n"); return 1; // Exit program due to invalid
        input
    }
}

// Switch Case Logic switch (choice) {
case 1:
    // Check if all digits are even temp = num;
    all_even = 1; // Assume true initially

    while (temp > 0) { rem = temp
        % 10;
        if (rem % 2 != 0) {
            all_even = 0; // Found an odd digit break;
        }
        temp /= 10;
    }

    printf("Number: %lld\n", num); if (all_even) {
        printf("Output: All digits are even\n");
    } else {
        printf("Output: All digits not even\n");
    }
    break;

case 2:
    // Check if all digits are prime temp = num;
    all_prime = 1; // Assume true initially while (temp > 0) {

```

```

        rem = temp % 10;
        // Prime digits are 2, 3, 5, 7
        if (rem != 2 && rem != 3 && rem != 5 && rem != 7) {
            all_prime = 0; // Found a non-prime digit
            break;
        }
        temp /= 10;
    }

    printf("Number: %lld\n", num);
    if (all_prime) {
        printf("Output: All digits are prime\n");
    } else {
        printf("Output: All digits are not prime\n");
    }
    break;

default:
    printf("Invalid choice. Please choose 1 or 2.\n");
    break;
}

return 0;
}

```